INCH-POUND
MIL-DTL-24643/18F
1 October 2009
SUPERSEDING
MIL-DTL-24643/18E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +105 °C, 1000 VOLTS, TYPE LSMSCU (INCLUDING VARIATION LSMSCS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight with Circuit Integrity

First	-	Copper conductor, coated or uncoated, 18 AWG, Class B stranding.
Second	-	Thermoset insulation. Minimum average wall thickness of 0.018 inch. Diameter of the insulation 0.084 inch minimum.
Third	-	Optional glass braid.
Fourth	-	Optional covering (unless required by 3.3.5 of MIL-DTL-24643). Diameter over the glass braid covering shall be a maximum of 0.113 inch. Standard identification code applied by Method 1.
Fifth	-	The required number of conductors (see table I or II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Except the lay of the two outer layers for sizes 44, 61, and 91 shall not exceed 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Sixth	-	An optional binder.

Construction, Variant Type LSMSCU

Seventh	-	Cross-linked polyolefin jacket (see table I or II for thickness).

Construction, Variant Type LSMSCS

Seventh	-	Braided shield of tin-coated copper.
Eighth	-	Spirally wrapped polyethylene terephthalate (polyester film) separator tape.
Ninth	-	Braided shield of tin-coated copper.
Tenth	-	An optional binder.
Eleventh	-	Cross-linked polyolefin jacket (see table I or II for thickness).

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TABLE I. Details (unshielded).

Military part no. M24643/18	Type and size	No. of conductors	Jacket thickness min. avg. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UN	LSMSCU-7	7	0.040	0.447	0.484
-02UN	LSMSCU-10	10	0.050	0.575	0.622
-03UN	LSMSCU-14	14	0.050	0.617	0.668
-04UN	LSMSCU-19	19	0.050	0.682	0.738
-05UN	LSMSCU-24	24	0.050	0.790	0.855
-06UN	LSMSCU-30	30	0.050	0.833	0.901
-07UN	LSMSCU-37	37	0.060	0.926	1.002
-08UN	LSMSCU-44	44	0.060	1.030	1.114
-09UN	LSMSCU-61	61	0.060	1.156	1.250
-10UN	LSMSCU-91	91	0.060	1.369	1.480

TABLE II. Details (shielded).

Military part no. M24643/18	Type and size	No. of conductors	Jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min. (inch)	max. (inch)
-01UD	LSMSCS-7	7	0.040	0.507	0.544
-02UD	LSMSCS-10	10	0.050	0.635	0.682
-03UD	LSMSCS-14	14	0.050	0.677	0.728
-04UD	LSMSCS-19	19	0.050	0.742	0.798
-05UD	LSMSCS-24	24	0.050	0.850	0.915
-06UD	LSMSCS-30	30	0.050	0.893	0.961
-07UD	LSMSCS-37	37	0.060	0.986	1.065
-08UD	LSMSCS-44	44	0.060	1.090	1.174
-09UD	LSMSCS-61	61	0.060	1.216	1.310
-10UD	LSMSCS-91	91	0.060	1.429	1.540

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.) 7.07

Voltage withstand (volts, root mean square, min.)

Conductor to conductor 3000

Conductor to shield (Type LSMSCS only) 1500

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Overall shield to overall shield (Type LSMSCS only)	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield (Type LSMSCS only)	500
Overall shield to overall shield (Type LSMSCS only)	100
Conductor continuity	No failure
Shield continuity (Type LSMSCS only)	No failure
Group A:	
Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (scrapes min.)	No damage
Crack resistance	No damage
Group B:	
Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Gas flame (1 hour)	No failure
NOTE: Type LSMSCS only, shields not grounded during test.	
Group C:	
Physicals (aged) air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60

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Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Surface transfer impedance (Type LSMSCS only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (minus 20±2 °C)	No damage
Gas flame (3 hour)	No failure
NOTE: Type LSMSCS only, shields not grounded during test.	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45

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Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSMSCU, LSMSCS	7 and 10	1500
All other sizes		1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-022)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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